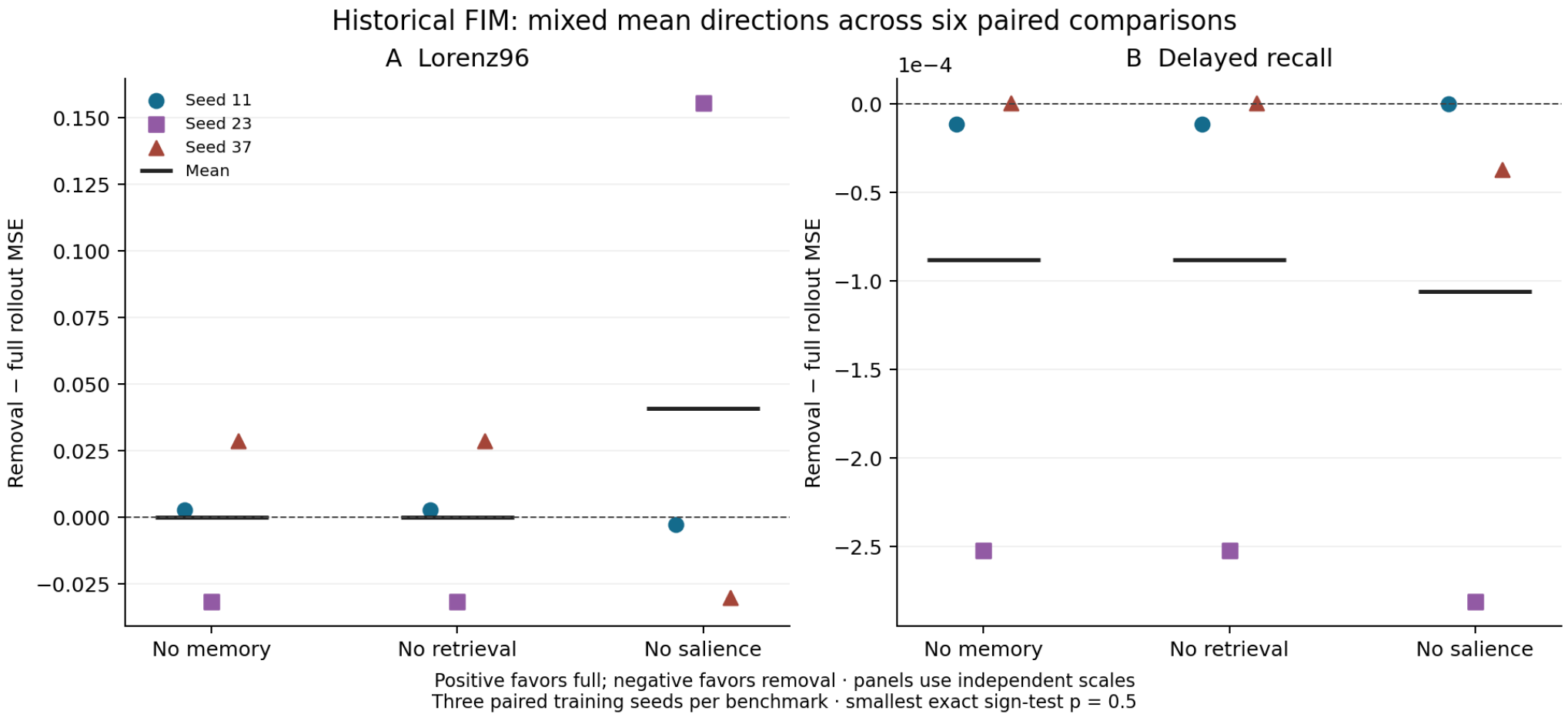


Retained component ablations in a historical three-seed FIM evaluation

Presentation draft. Author list and affiliations require review.

Mean directions split 3-3 across six comparisons. The smallest exact sign-test p-value is 0.5.



All paired removal-minus-full rollout-MSE differences are shown. Positive favors the full model; negative favors removal. The two benchmark panels have independent vertical scales.

Retained design and audit

Two benchmarks × three seeds × four configurations give 24 retained cells. Configurations are full, no memory, no retrieval and no salience gating. The shared-minibatch protocol used 12 epochs, batch size 64 and four-step training versus 30-step evaluation rollouts.

All 96 text-artifact hashes were checked: a summary, run-results report, configuration and log for each cell. MSE and MAE agree across the retained reports and manifest. No checkpoint was read or replayed.

Seed-level interpretation

The largest mean full-model advantage, Lorenz96 salience gating, is driven by seed 23; the other two seeds favor removal. No-memory and no-retrieval error sequences coincide in both benchmarks.

The result remains negative

The retained comparisons do not establish component necessity or a recovered memory mechanism. Matching error sequences do not prove prediction or implementation identity. Three seeds do not supply independent-domain replication.

Statistical and source limits

With three nonzero pairs, the smallest possible two-sided exact sign-test p-value is 0.25; the observed minimum is 0.5. Nonsignificance does not establish equivalence. The executed Git revision is recorded as UNKNOWN, so full source reconstruction is not claimed.

A distinct historical lineage

This presentation covers the closed historical shared-minibatch study. Burgers V2, delayed-recall V2 and the held trajectory-isolated successor are separate projects. Their sources and outcomes are not substituted by this package.